# A Framework for Emergent Time from Horn-Filling Dynamics

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**Abstract**

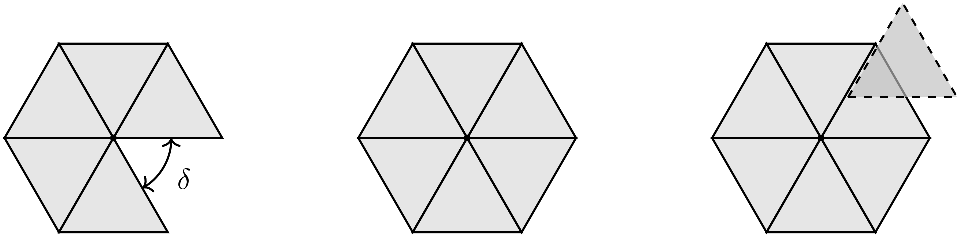
We propose a framework in which spacetime is a growing simplicial set and time is the process of filling its inner horns. Two local, non-deleting moves drive the growth of spacetime: a "sprout" attaches a fresh directed edge ending at a new vertex, and a "fill" completes an unfilled inner horn with a composite edge and its witnessing cell. Each move is a weak homotopy equivalence of the underlying space, so the homotopy type is conserved at every step. Since the geometric realization of the simplicial set forgets edge orientation, this conserved type cannot see any arrow of time; all physical content lives in the directed structure that realization forgets. The completed limit of the filling process is a quasicategory, in which every composition question has an answer while edges remain directed. Each move is an event that consumes cells created by earlier events, the dependency of moves forms a partial order, and a reference frame is a choice of linearization of this order. The entropy of an event is the logarithm of the number of linear extensions of its causal past. This quantity measures how much of the past could have happened in a different order, and grows monotonically, which we prove; a toy implementation provides first numerical support. The arrow of time appears as a broken symmetry. The universe carries an arrow exactly when it is not isomorphic to its own edge-reversal, and pairing the universe with its mirror image restores the symmetry globally, in the spirit of CPT-symmetric cosmologies. We formulate a maximum-entropy inference problem whose solution, conjecturally of Gibbs form, would determine the rule from locality and a fixed update budget alone. We close with the open problems, foremost the recovery of the Lorentz group and of an area law for horizon entropy.

## 1 Introduction

Discrete approaches to quantum gravity, from Regge calculus [22] to causal dynamical triangulations [3], replace smooth spacetime with combinatorial data: simplices, gluings, and the lengths assigned to them. In Regge calculus the combinatorial scaffold is fixed with only edge lengths allowed to vary. The vertices, simplices, and gluings of the triangulation never change, so no new combinatorial structure is ever introduced. That is enough for classical gravity, where one solves the field equations on a triangulation that is given in advance. The discomfort begins one step later, when Feynman's path integral is applied to the evolution of the universe and the sum runs over all histories. Summing over Regge geometries means integrating over edge lengths, a continuous space in which many assignments describe the same geometry and no canonical measure presents itself. On the other hand, dynamical triangulations [11, 1, 2] escape this by inverting Regge's choice. The lengths are frozen and the combinatorics is set free, so a geometry becomes a gluing pattern of simplices whose shapes are fixed once and for all, and summing over geometries becomes counting gluings.

In both cases the geometry is carried by the gluing and not by any ambient space. Each simplex is flat inside, so curvature can only sit where simplices meet, and it is measured by how far the angles around such a meeting place fall short of a full turn. Figure 1 shows the three cases for triangles glued around a vertex.

Because the combinatorics is now free, the number of simplices differs from history to history, and within a single history the number of simplices in a slice changes from one slice to the next.




$k = 5$, angles sum to 300° gap of 60°, cones up

$k = 6$, angles sum to 360° no gap, stays flat

$k = 7$, angles sum to 420° excess of 60°, buckles

Figure 1: Flat equilateral triangles glued around a single vertex. With five triangles the angles at the vertex sum to 300°, the surface cannot lie flat, and it cones up. With six they sum to 360° and the surface is genuinely flat. With seven they sum to 420°, there is no room for the seventh triangle in the plane, and the surface buckles into a saddle. The shortfall $\delta = 360° - \sum_i \theta_i$ is called the deficit angle, and it accounts for all of the curvature of the glued surface.

New geometric measure appears from nowhere. Classical general relativity is no better off on this point, since it has no conservation law for spatial volume either, and in a closed universe the Hamiltonian constraint balances the total at zero [4, 12] however much volume appears. The complaint is therefore not that volume grows, but that the growth answers to nothing the theory conserves. A cosmological constant term only assigns a cost to four-volume within the ensemble, and where the total is held fixed instead, as in simulations and in unimodular gravity [26, 15], the constraint is imposed by hand rather than descending from a symmetry.

This paper takes the discomfort seriously and asks what a theory looks like if one demands, from the start, that

1. Nothing is ever destroyed. Evolution proceeds only by forking and completion moves, never by deletion, so that topological invariants are conserved by construction.

2. Time is not a dimension of the fundamental structure but an emergent bookkeeping of an irreversible forking-filling process.

3. Entropy is attached to frame-invariant causal data, not to spatial slices, so that the theory never needs to say what all of space looks like at a single moment.

4. The rule that chooses which move happens next is derived from an inference principle rather than postulated.

In our framework the universe begins as a single vertex, which we call the big bang, and grows by two kinds of local moves:

1. A **sprout** attaches a fresh directed edge ending at a new vertex, adding new space.

2. A **fill** completes an unfilled inner horn, attaching a composite edge together with the higher cell that witnesses the composition.

Two kinds of direction appear here. Every edge of the simplicial set carries an orientation, and that orientation is what decides which horns count as inner and what composition means. Separately, every move is an event, and the events are ordered by which of them created the cells the others consumed. We call the first the **orientation** and the second the **causal order**. No move ever deletes anything, so growth always runs forward in the causal order, while orientation is a property of the finished cells and can be reversed on paper.

At the big bang two edges sprout from the single vertex with opposite orientation, and this is what produces the two branches. The mirror branch is not an independent run. By definition it is the edge-reversal of the branch, carrying one mirrored event for every event, which is what makes the pair a mirror image of itself. The mirroring is causally inert, since no move in one branch ever consumes a cell created in the other, so the causal past of any event lies wholly inside its own branch. Each branch grows forward in its own causal order while pointing the opposite way in orientation, so neither branch runs backward in time in any absolute sense. Each is backward only as seen from the other, and there is no global causal order in which either of them is reversed. This is a combinatorial counterpart of CPT-symmetric cosmologies, where a universe and an antiuniverse emerge together from a single point [7, 5].

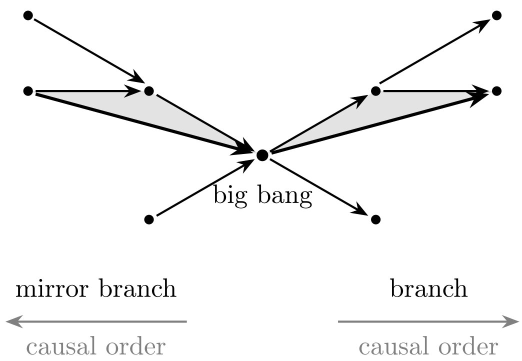



Figure 2: The total object. A branch on the right and its mirror image on the left, glued at the single vertex where both were born. Corresponding edges carry opposite orientation, so the pair is isomorphic to its own edge-reversal. The shaded region on each side is a filled inner horn, drawn with its composite edge in bold, and the fill is mirrored along with everything else. The grey arrows are the causal order, which runs away from the shared vertex on both sides, so each branch is backward only as seen from the other, and never in the order in which its own cells were built.

At any moment the state of the universe is a finite simplicial set with directed edges, together with the history of how it was built. The limit of the filling process in one branch is a quasicategory, in which every composition has been formed while the orientations remain. The classical, zero-information limit is the nerve of an ordinary category, the simplicial set whose $n$-cells are the chains of $n$ composable morphisms [24, 19], and in it every filler is unique. The physical content of a completed universe is precisely how far it is from being a nerve.

The pair of branches is meant to carry no information at all. Two contractible halves glued at a single vertex are again contractible, so every homotopy invariant of the pair is that of a point, and the pair is isomorphic to its own edge-reversal, so it carries no arrow either. Whatever is physical therefore sits inside a single branch, in the orientations and in the history of how its cells were built.

# 2 Preliminaries and related work

The first three subsections fix every notion used later, bottom-up, so that no concept is used before it is defined. The last places the framework against existing work.

## 2.1 Simplicial sets, horns, and quasicategories

**Definition 2.1** (Simplicial set)**.** Let $\Delta$ denote the category whose objects are the finite nonempty linear orders $[n] = \{0 < 1 < \cdots < n\}$ and whose morphisms are order-preserving maps. A *simplicial set* is a functor $X : \Delta^{\mathrm{op}} \to \mathsf{Set}$. The set $X_n = X([n])$ is the set of *$n$-simplices*; $X_0$ are *vertices*, $X_1$ are (directed) *edges*. The category of simplicial sets is denoted $\mathsf{sSet}$.

**Example 2.2.** The *standard $n$-simplex* $\Delta^n = \Delta(-, [n])$. Concretely, $\Delta^2$ has three vertices $0, 1, 2$, three edges $0 \to 1$, $1 \to 2$, $0 \to 2$, and one nondegenerate 2-simplex filling the triangle.

**Definition 2.3** (Horn)**.** For $0 \leq k \leq n$, the *horn* $\Lambda^n_k \subset \Delta^n$ is the subcomplex obtained by removing the interior of $\Delta^n$ and the face opposite vertex $k$. A horn is *inner* if $0 < k < n$.

**Example 2.4.** $\Lambda^2_1$ consists of two composable edges $0 \to 1 \to 2$ with the composite $0 \to 2$ and the filling triangle *missing*. Filling $\Lambda^2_1$ means supplying both.

**Definition 2.5** (Kan complex; quasicategory; nerve)**.** A simplicial set $X$ is a *Kan complex* if every map $\Lambda^n_k \to X$ extends along $\Lambda^n_k \hookrightarrow \Delta^n$ for all horns, and a *quasicategory* if this holds for all *inner* horns. The *nerve* $\mathrm{N}(\mathcal{C})$ of a small category $\mathcal{C}$ is the simplicial set with $\mathrm{N}(\mathcal{C})_n =$ chains of $n$ composable morphisms. A simplicial set is a nerve of a category iff every inner horn has a *unique* filler [19].

*Remark* 2.6 (The spectrum used in this paper). Bare directed graphs (only 0- and 1-cells): nothing filled. Quasicategories: everything filled, fillers unique only up to higher cells. Nerves: everything filled uniquely. We will interpret these respectively as the newborn universe, the completed universe, and the classical (zero-information) limit.

*Remark* 2.7 (Opposites). The involution of $\Delta$ reversing each linear order induces $X \mapsto X^{\mathrm{op}}$, under which $\Lambda^n_k$ becomes $\Lambda^n_{n-k}$. The two outer horns are exchanged and inner horns go to inner horns, so $X$ is a quasicategory exactly when $X^{\mathrm{op}}$ is. Reversing orientations therefore cannot turn an inner-horn filling policy into an outer-horn one.

### 2.2 Rewriting, events, and causal order

**Definition 2.8** (Non-deleting rewriting rule)**.** A *rule* is a monomorphism $L \hookrightarrow R$ of finite simplicial sets. An *application* of the rule to a configuration $X$ is a choice of match $m : L \to X$; the result is the pushout $X \cup_L R$. Because $L \hookrightarrow R$ is a monomorphism and nothing is removed, applications only add cells. (This is the non-deleting special case of double-pushout graph rewriting [13], where the interface equals the left-hand side.)

**Definition 2.9** (Run; event; causal order)**.** A *run* is a finite sequence of rule applications starting from an initial object. Each application is an *event*. Event $b$ *depends* on event $a$ (written $a \prec b$) if the match of $b$ uses a cell created by $a$; the reflexive-transitive closure $\leq$ of $\prec$ is the *causal order*. The *causal past* of an event $e$ is $J^-(e) = \{a : a \leq e\}$.

*Remark* 2.10 (Orientation is not the causal order). Two directions must be kept apart. The *orientation* of an edge is part of the simplicial set and decides which horns are inner. The causal order of Definition 2.9 is an order on events and records which move supplied the cells another move consumed. Since no rule deletes anything, the causal order is a partial order whatever the orientations do, and a branch whose edges all point towards its origin still grows away from it.

*Remark* 2.11. The triple (events, causality, independence) is an event structure in the sense of Winskel [27]; we use only the partial order.

**Definition 2.12** (Linearization; frame)**.** A *linearization* (or *linear extension*) of a finite poset $(P, \leq)$ is a total order on $P$ extending $\leq$. We write $\mathrm{Ext}(P)$ for the number of linear extensions. In this framework a *reference frame* is a linearization of the causal order.

### 2.3 Associahedra and the coherence tower

**Definition 2.13** (Catalan numbers; associahedron)**.** The $n$-th Catalan number is $C_n = \binom{2n}{n}/(n+1)$; $C_{n-1}$ counts the full bracketings of $n$ composable edges. The *associahedron* $K_n$ [25] is the polytope whose vertices are these bracketings and whose faces encode partial bracketings; $K_4$ is the pentagon.

The relevance is dynamical: filling 2-horns creates composites, whose alternative bracketings pose 3-horn questions, whose answers pose compatibility (4-horn, pentagon) questions, and so on. Completion promotes questions up this tower; it never depletes them. Since the number of bracketings grows like $C_{n-1} \sim 4^n/n^{3/2}$, question supply grows exponentially with tower level.

### 2.4 Related work

**[GAP: This subsection must be revised after the adversarial literature scan is complete. Overlap verdicts below are provisional.]**

**Discrete gravity.** Regge [22] showed that curvature on a piecewise-flat complex is concentrated on hinges as deficit angles and that the corresponding action converges to the Einstein–Hilbert action. Dynamical triangulations freeze edge lengths and make the gluing combinatorics dynamical; causal dynamical triangulations [3] additionally impose a global time foliation and recover a de Sitter-like volume profile numerically. Our framework shares the combinatorial ontology but differs in three respects: moves are non-deleting (CDT's Monte Carlo moves are not), no foliation is imposed (causal order is derived from move dependency), and the object grown is a directed simplicial set headed toward a quasicategory rather than a triangulated pseudomanifold of fixed topology.

**Causal sets.** Causal set theory [6] takes a locally finite partial order as the fundamental ontology, with order-plus-number determining geometry, and classical sequential growth [23] supplies a stochastic birth process. Our causal order is likewise fundamental-but-derived (from move dependency), and our sprouting move is a birth event; the essential difference is that our elements are not featureless points but cells of a simplicial set, so that composition, coherence, and the entropy of Section 3.3 have combinatorial substrate. To our knowledge the linear-extension entropy of a causal past has not been used as a dynamical functional in the causal set literature. **[GAP: verify this claim in the lit scan]**

**Rewriting cosmologies.** The Wolfram physics project [14] evolves hypergraphs by rewriting rules and argues that *causal invariance* (confluence of the rewriting system) yields observer-independence of the causal graph and, conjecturally, Lorentz covariance. We adopt the same event/causal-graph architecture and the same open problem (Section 5.2); our divergences are the specific simplicial move set (horn filling, with the quasicategory as limit object), the derived rather than postulated rule (Section 3.5), and the entropy functional.

**Emergent and thermal time.** That time may be state-dependent is the thermal time hypothesis of Connes and Rovelli [10], built on Tomita–Takesaki modular flow; that time may be entanglement is Page and Wootters [21]. Our proposal is coarser but of the same family: time is identified with an irreversible completion process, and equilibrium (the quasicategory limit, no unfilled inner horns) is the state in which the process halts.

**CPT-symmetric cosmology.** Pairing a universe with its mirror image so that the total object carries no arrow is the combinatorial counterpart of the universe-antiuniverse pair of Boyle et al. [7] and of the two arrows emerging from a shared middle in the Janus point picture of Barbour et al. [5]. In both of those the mirror branch is determined by the branch rather than evolving independently, which is also our convention.

**Entropic derivations.** Jaynes' maximum-entropy inference [17], its dynamical extension by Caticha [9], and Jacobson's derivation of the Einstein equations from the Clausius relation on local causal horizons [16] together define the strategy of Sections 3.5 and 5.2: derive the rule by inference, and aim the long game at an area law.

**Computation-budget kinematics.** The Margolus–Levitin bound [20] caps the rate of distinguishable state changes by energy, and Lloyd [18] applies it cosmologically. Our budget axiom (Section 3.4) is of this lineage.

# 3 The framework

## 3.1 Axioms

**Axiom 1** (Ontology)**.** The state of the universe is a finite simplicial set with directed edges, together with its construction history. History is primary; the spatial state is derived (Definition 3.3).

**Axiom 2** (Dynamics)**.** Evolution proceeds by two local, non-deleting moves: *sprout* and *fill* (Definition 3.1). Only inner horns may be filled.

**Axiom 3** (Budget)**.** The total rate of moves is fixed. The constant $c$ is the exchange rate between translation and internal solving (Section 3.4).

**Axiom 4** (Inference)**.** The probability distribution over available moves is the maximum-entropy distribution consistent with Axioms 2–3 and locality (Section 3.5). No further dynamical input is allowed.

## 3.2 State space and moves

**Definition 3.1** (Moves)**.** Fix the two rules, as monomorphisms of finite simplicial sets:

1. **Sprout:** $\Delta^0 \hookrightarrow \Delta^1$. A match is a vertex $v$; the application attaches a fresh directed edge from $v$ to a new vertex.
2. **Fill:** $\Lambda^n_k \hookrightarrow \Delta^n$ for $0 < k < n$. A match is an unfilled inner horn; the application attaches the missing face and interior. Each horn may be filled at most once per distinct match. **[GAP: decide and justify the at-most-once convention versus allowing multiple fillers of the same horn; interacts with the choice-entropy variant of Section 5.1]**

The initial object is a single vertex $\Delta^0$ (the big-bang event).

*Remark* 3.2 (Conservation for free). Sprouting is a Whitehead elementary expansion and inner-horn filling is an anodyne extension; both are homotopy equivalences. Hence the classical homotopy type of the universe is conserved by construction; indeed, it remains that of a point. All evolving information is therefore *directed* structure, invisible to classical homotopy invariants. This is a feature: the framework's physics is located strictly below classical homotopy equivalence, in composition and coherence data.

**Definition 3.3** (State as colimit of the causal past)**.** Let $e$ be an event of a run, with causal past $J^-(e)$ (Definition 2.9). The *state at $e$* is the colimit of the diagram of rule applications in $J^-(e)$. Since colimits depend only on the diagram and not on any ordering of it, the state at $e$ is independent of the choice of linearization: no foliation enters.

**Proposition 3.4** (Well-definedness)**.** *The dependency relation of Definition 2.9 is a partial order, and the colimit in Definition 3.3 exists and is independent of linearization.*

**[GAP: Proof to be written (target: next session). Sketch: antisymmetry from acyclicity of creation-before-use; colimit exists since sSet is cocomplete; independence is the universal property.]**

### 3.3 The entropy functional

**Definition 3.5** (Causal-past entropy)**.** For an event $e$ of a run,

$$S(e) \;=\; \log \mathrm{Ext}\big(J^{-}(e)\big),$$

the logarithm of the number of linear extensions of its causal past.

The coarse-graining implicit in $S$ is canonical to the framework: the state (colimit) forgets the order of causally independent events, and $S$ counts exactly the forgotten orderings. A chain has $S = 0$; an antichain of $n$ events has $S = \log n!$. Entropy measures causal parallelism.

**Lemma 3.6** (Monotonicity)**.** *If $e \le e'$ then $S(e) \le S(e')$.*

**[GAP: Proof to be written (owner: ixaxaar; the injection of extension sets should be a half-page). Empirically verified in Section 4.]**

**Lemma 3.7** (Foliation invariance)**.** *$S(e)$ depends only on the isomorphism class of the poset $J^{-}(e)$ and in particular is independent of any linearization of the run.*

**[GAP: Proof to be written; content is the frame-independence of $J^{-}(e)$.]**

*Remark* 3.8 (Cost of computation). Counting linear extensions is #P-complete in general [8]; Section 4 uses exact counting on small runs and future work requires a sampling estimator.

### 3.4 Kinematics: budget, proper time, mass

Under Axiom 3 the total move rate available to any subsystem is fixed. For a subsystem in uniform translation at speed $v$, the budget splits between translation and internal solving; taking the Euclidean split of the light-clock argument, the internal rate is $c\sqrt{1 - v^2/c^2}$, reproducing time dilation. Proper time along a worldline is the accumulated internal solving; a null worldline solves nothing (a photon has no internal clock), and, conversely, any system with internal evolution (an oscillation, a tick) must be massive. We identify mass with the rest-frame solving rate, in the spirit of the Compton clock $mc^2/\hbar$.

Two honest limitations. First, uniform motion must be locally undetectable: all internal processes of a moving subsystem slow *coherently*, so rate changes are visible only upon reunion; any locally measurable rate change would falsify the framework immediately. Second, the budget argument yields dilation only. It does not by itself yield relativity of simultaneity or the Lorentz group; that requires the confluence programme of Section 5.2.

### 3.5 The maximum-entropy problem

**Definition 3.9** (Move ensemble)**.** At a given state, let $M = M_{\mathrm{sprout}} \sqcup M_{\mathrm{fill}}$ be the set of available moves (legal sprout sites and unfilled inner horns), and let $k : M \to \mathbb{R}_{>0}$ be a cost function depending only on a move's level $d$ in the coherence tower.

**Definition 3.10** (The inference problem)**.** Find the probability distribution $q$ on $M$ maximizing the Shannon entropy $H(q) = -\sum_m q(m) \log q(m)$ subject to

(C1) normalization, $\sum_m q(m) = 1$;

(C2) budget, $\mathbb{E}_q[k] = R$;

(C3) locality and homogeneity: $q(m)$ depends only on the isomorphism class of the local neighbourhood of $m$'s match.

**Conjecture 3.11** (Gibbs rule)**.** *The problem of Definition 3.10 has a unique solution of the form $q(m) \propto e^{-\lambda k(m)}$ with a single multiplier $\lambda(R)$. The sprout-to-fill ratio and the expansion law $N(t)$ are then determined by the level degeneracies (Catalan counts) and $k$, with no further choices.*

**[GAP: Derivation is the subject of a dedicated session. Danger to watch: whether (C3) can be formulated without smuggling in a preferred scale.]**

### 3.6 Coherence escalation and decelerating expansion

Filling promotes questions up the associahedral tower (Section 2): solving at level $d$ mints questions at level $d+1$, with supply growing like $4^d$. Under Axiom 3, if the cost of a solve grows with level, the solve rate obeys

$$\frac{dN}{dt} = \frac{R}{k\big(d(t)\big)},$$

and since consuming level $d$ takes time exponential in $d$, the typical level grows slowly, $d(t) \sim \log t$, forcing sub-linear, power-law-like growth of $N(t)$. The cost function $k$ plays the role of matter content: different $k$ give different deceleration exponents. **[GAP: Make this a theorem for at least one explicit $k$; compare exponents against the toy simulation once the sampling estimator allows long runs.]**

*Remark* 3.12 (Two arrows). The framework contains two independent monotone quantities: the entropy $S$ (Lemma 3.6) and the tower level $d$. That they are aligned is expected but unproved. **[GAP: prove or refute alignment]**

## 4 Numerical results

We implemented the dynamics of Definition 3.1 as a stochastic process on directed graphs (code in Appendix B; the implementation currently realizes sprouts and $\Lambda^2_1$ fills, i.e. tower level $d \leq 2$). At each tick a sprout occurs with probability $p_{\text{sprout}}$, otherwise a uniformly random unfilled inner horn is filled. Dependencies follow Definition 2.9: a sprout at $v$ depends on the event that created $v$; a fill of the horn $(u \to v \to w)$ depends on the events that created its two edges. The entropy of Definition 3.5 is computed exactly by dynamic programming over downsets of the event poset, which limits runs to $\approx 20$ events (the #P wall).

Three observations, with their status:

1. **Monotonicity** (supports Lemma 3.6): $S(t)$ is strictly increasing in every run.
2. **Ratio ordering**: $S(t)$ is pointwise increasing in $p_{\text{sprout}}$. Interpretation: expansion produces entropy; equilibration (filling) suppresses its growth. This links the entropy functional to the expansion dynamics and is the empirical hook for treating the dynamics as an entropic gradient flow. **[GAP: significance analysis: more seeds, error bars, and a test of pointwise ordering]**
3. **No visible deceleration**: growth is roughly linear at this scale, as expected, since 18 events barely populates tower level $d = 3$, so the mechanism of Section 3.6 cannot yet bite. **[GAP: sampling-based estimator for Ext (e.g. a Karzanov–Khachiyan-style Markov chain) to reach $10^2$–$10^3$ events, plus implementation of $d \geq 3$ horn filling, then test for power-law $N(t)$ and for entropy deceleration]**

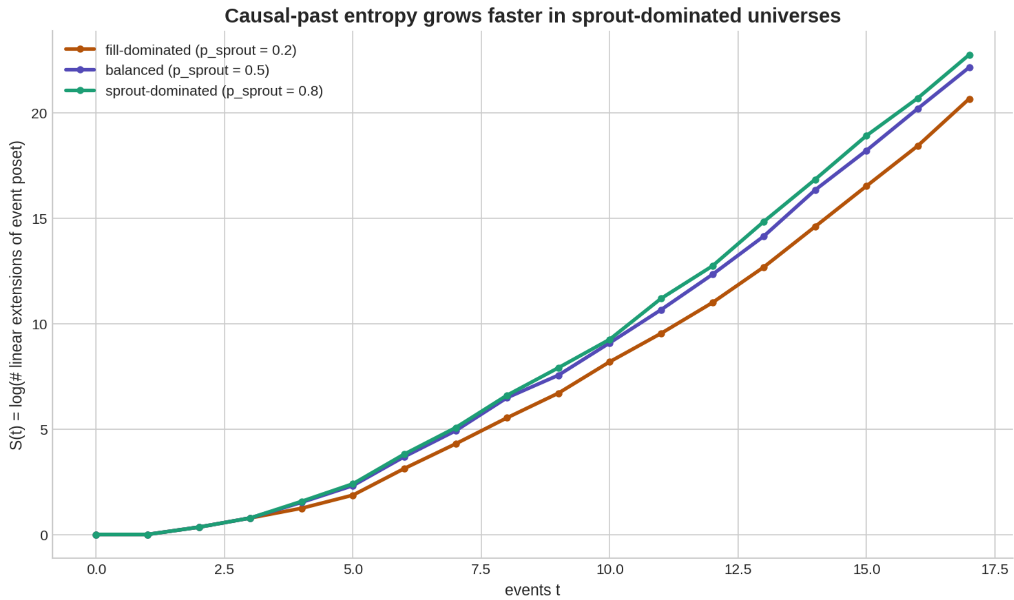



Figure 3: Causal-past entropy $S(t)$ (mean over six seeds, 18 events) for three sprout probabilities. Entropy is monotone in every run and ordered by the sprout fraction at every time: sprouts create causally independent events (parallelism), fills create dependencies (sequentialization).

# 5 Discussion

## 5.1 The design space

Every framework paper hides choices. This section exposes ours, records the alternatives, and states why we chose as we did, so that each rejected branch remains a well-posed future project.

**D1: What the cells are.** Chosen: simplicial sets with directed edges. Alternatives: (a) bare undirected graphs, the simplest option, but no home for composition or coherence data; (b) oriented simplicial complexes with group-labelled edges, the minimal setting where torsion-type invariants (Reidemeister/Whitehead) are nonzero, attractive because torsion is precisely the information surviving below homotopy equivalence, rejected for v1 on grounds of economy; (c) Kan complexes as states, rejected because a Kan complex is a *finished* object (every horn filled): it is the natural *endpoint* of the dynamics, not its carrier.

**D2: Which horns may be filled.** Chosen: inner horns only; the limit object is a quasi-category. Alternative: all horns (Kan limit), giving an undirected, fully invertible structure. Rejected because directedness provides native causal orientation, keeps inverses costly rather than free, and makes orientation-reversal (the Feynman–Stückelberg reading of antimatter) expressible. Note the arrow of time does *not* depend on this choice (it comes from non-deletion), so the Kan variant remains a coherent, reversible-substrate cousin worth exploring.

**D3: The creation move.** Chosen: sprouting from existing vertices. Alternatives: (a) a rich initial seed, rejected as engineered; (b) free appearance of disconnected cells, rejected as it breaks locality and connectedness. Sprouting preserves connectedness and the homotopy type (elementary expansion) by construction.

**D4: The trigger rule.** Chosen: none; the rule is to be derived by maximum entropy (Axiom 4, Conjecture 3.11). The postulated alternatives, each a different cosmology, are recorded for comparison: free sprouting (exponential, pure cosmological constant); frustration-triggered (self-limiting); fill-triggered (linear, expansion locked to solving rate); curvature-coupled via discrete Ricci curvature (geometry back-reaction, Einstein-equation-shaped). Under the max-ent programme these should reappear as consequences of which constraints are imposed.

**D5: What entropy counts.** Chosen (v1): orderings only, $S = \log \mathrm{Ext}(J^-(e))$. Alternative: orderings *plus* filler choices, giving a two-component entropy whose second term measures distance from nervehood, i.e. exactly the quasicategory-vs-nerve gap where we have argued the physics lives. Deferred, not rejected: the ordering entropy is the right first theorem target, and the choice term requires the multiple-filler convention of Definition 3.1 to be settled first.

## 5.2 Open problems

In decreasing order of severity.

**O1: The Lorentz group.** The budget argument yields time dilation; it does not yield relativity of simultaneity. The required property is causal invariance: confluence of the rewriting system, so that all linearizations generate the same causal order and no experiment detects a preferred one. Whether confluence plus causal structure yields the actual Lorentz group is open here as it is in the hypergraph-rewriting programme [14]. A specific sub-problem: the framework predicts a fact of the matter about the causal past of every event but must *not* predict a fact of the matter about "the total entropy of the universe now"; our slice-free entropy (Lemma 3.7) is designed for exactly this, but the claim needs a theorem, not a design intention.

**O2: The area law.** The long-game target is Jacobson's derivation [16]: if the path entropy of a causal horizon is proportional to its *area*, the Einstein equations follow as an equation of state. If it is proportional to volume, the framework is dead as a theory of gravity. Computing the scaling of $S$ across a causal horizon in the toy model is therefore the single most falsifying computation available to us. **[GAP: define "horizon" and "area" internally first]**

**O3: Emergent unitarity.** The substrate is irreversible; the effective dynamics encoded in the record must nevertheless be unitary and CPT-symmetric. The ledger-of-reversible-moves picture is a metaphor until an effective Hilbert-space dynamics is extracted; the difficulty is of the same kind as in the cellular-automaton interpretation of quantum mechanics.

**O4: Geometry.** Combinatorial volume is not metric geometry. Distances, dimension (spectral or Hausdorff), and curvature must be shown to emerge; discrete Ricci curvatures and spectral-dimension flow are the natural instruments, and comparison with the CDT dimensional-reduction results [3] is an obvious benchmark.

**O5: The conserved charge of the tower.** Solving promotes questions up the coherence tower monotonically. Whether a genuine conserved flux ("question current") exists, and whether the two arrows ($S$ and $d$) are provably aligned, is open.

## 5.3 Conclusion

We have proposed a framework in which space is a growing directed simplicial set, time is the process of answering its composition questions, entropy is the count of forgotten orderings of a frame-invariant causal past, and the dynamical law is an inference, not a postulate. The proved core is small, one experiment exists, and the failure modes are stated. The next steps are fixed: the two lemma proofs, the max-ent derivation, the sampling estimator, and the literature scan, in that order of weekends.

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# A The budget triangle

Let the total move rate of any subsystem be $c$ (Axiom 3) and suppose a subsystem translates at speed $v$. Writing the split as orthogonal components, the internal rate $r$ satisfies $r^2 + v^2 = c^2$, hence

$$r \;=\; c\sqrt{1 - v^2/c^2},$$

the standard dilation factor, with the photon limit $r = 0$ at $v = c$. The orthogonal (Euclidean) composition is here an *assumption*; the question of why the emergent geometry of linearizations is Lorentzian rather than Euclidean, that is, why straight worldlines *maximize* accumulated solving, is part of problem O1.

# B Simulation code

The toy implementation (also shipped as `bin/toy_universe.py`): sprout and $\Lambda^2_1$-fill dynamics, event poset construction per Definition 2.9, and exact linear-extension counting by dynamic programming over downsets.

```
class Universe:
    # vertices, directed edges, unfilled inner horns;
    # every move is an event with explicit parent events.

    def sprout(self):
        # new vertex w, new edge v -> w
        # parents = { creator_event(v) }
        ...

    def fill(self, horn):
        # horn = (e1: u -> v, e2: v -> w)
        # new composite edge u -> w (+ 2-cell)
        # parents = { creator(e1), creator(e2) }
        ...
```

```
def count_linear_extensions(parents):
    # DP over downsets of the event poset (bitmask),
    # processing masks in order of popcount.
    ...
```

Full listing in the repository. Runs: 18 events, six seeds, $p_{\text{sprout}} \in \{0.2, 0.5, 0.8\}$; output reproduced in Figure 3.